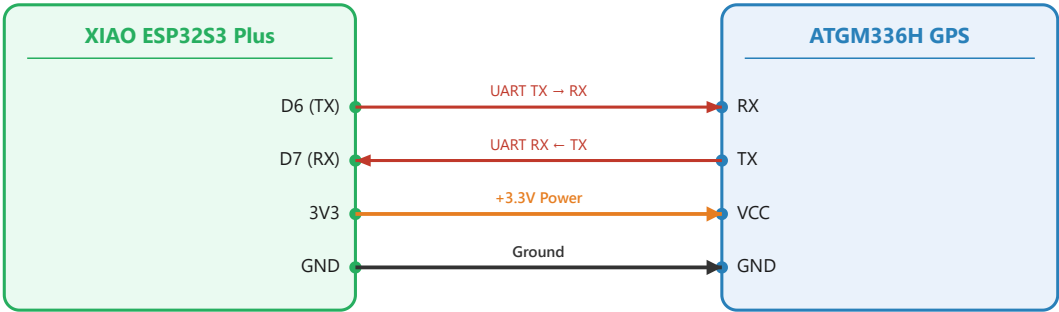


RadiaLog — Wiring Diagram

XIAO ESP32S3 Plus · ATGM336H GPS · 18650 Battery Voltage Divider

1. GPS Module (ATGM336H)

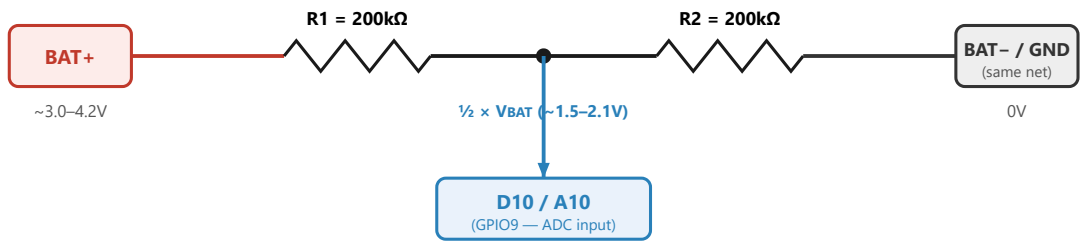


XIAO Pin (silkscreen)	GPIO	Direction	GPS Pin	Purpose
D6 (TX)	GPIO43	→	RX	UART data to GPS
D7 (RX)	GPIO44	←	TX	UART data from GPS (NMEA)
3V3	—	→	VCC	Module power
GND	—	—	GND	Common ground

GPS sleep is handled in software. Before entering deep sleep, the firmware sends the ATGM336H a `$PMTK161,0` backup-mode command over UART, which drops its draw from ~25 mA to roughly 10–50 μ A while VCC stays applied. No hardware power-control pin is required.

UART data +3.3V power Ground

2. Battery Voltage Divider



- Two matching 100kΩ through 220 kΩ resistors in series between BAT+ and GND halve the battery voltage.
- Midpoint tap feeds D10 / A10 (GPIO9), within the ESP32 ADC range.
- High resistance = negligible drain (~10 μA).
- BAT- is the same net as GND — you can run the divider directly across the battery terminals.

3. GPS Power States at a Glance

State	ATGM336H	Approx. current
Normal run (tracking)	ON	~25 mA
Deep sleep / shipping mode (after \$PMTK161,0)	Backup	~10–50 μA

Source: README.md — branch feat/gps-power-control . Print to PDF via your browser (Ctrl/Cmd + P → Save as PDF).